The communication protocol for set the controller ID, baud rate and network parameters

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1.1 General agreement of communication

Data packet is used to communicate between the PC and the controller, in order to enhance the reliability of data, expanding capabilities to deal with images and other data.

Communication process：

a).PC send a data packet to the controller；

b).The controller received the data packet, analysis and processing of data packet, and then return a data packet to PC if necessary；

c). PC receives the data packets returned from the control card, and analysis the received data packets to determine whether communication is successful.

Serial setting:

Baud rate: 115200, 57600, 38400, etc. selected by the selected tool.

Format string: "115200, 8, N, 1", you can change the baud rate value 115200 according to what you set to the controller.

Packet data checksum

Communication process using the packet data checksum to check the correctness of data transmission, Checksum calculations we should pay attention：Data checksum is cumulative each byte of data, use the 16bit (2 bytes) unsigned number to represent, so when the data validation is more than 0xFFFF, the checksum, and retain only 16bit value. For example, 0xFFFA + 0x09 = 0x0003.

2 Data packet format

2.1 RS232/RS485 data packet format

2.1.1 The data package constitutes

Sent packets and return packets have adopted the following packet format：

Data	Value	Length(Byte)	Description
Start code	0xa5	1	The start of a packet
Packet type	0x68/0xE8	1	Recognition of this type of packet Send packet： 0x68 Return packet： 0xE8
Card type	0x32	1	Fixed type code
Card ID	0x01~0xFE 0xFF	1	Control card ID, the screen No, valid values are as follows: 1 ~ 254: the specified card ID 0XFF: that group address, unconditionally receiving data
Command code (CMD)	See Command list	1	To perform the specified command
Additional information/ confirmation mark	0或1	1	The meaning of bytes in the packet is sent, "Additional Information", is a packet plus instructions, and now only use the lowest: bit 0: whether to return a confirmation, 1 to return and 0 not to return bit1 ~ bi7: reserved, set to 0
Packet data	CC 。。。。。。	Variable-length	Data
Packet data checksum	0x0000~0xffff	2	Two bytes, checksum。 Lower byte in the former。The sum of each byte from " Packet type " to " Packet data" content
End code	0xae	1	The end of a packet（Package tail）

2.1.2 RS232/RS485 packet data transcoding description:

The following process is done sending and receiving low-level functions, If you write your own PC side of the sending and receiving programs, you must implementation as below conventions. Use the without transcoding code to calculation checksum.

Send :
Between start code and end code, if there is 0xA5, 0xAA or 0xAE, it should be converted to two code :

0xa5 ➔ 0xaa 0x05。 The purpose is to avoid the same with the start character 0xa5

0xae ➔ 0xaa 0x0e。 The purpose is to avoid the same with the end of the symbol 0xae。

0xaa ➔ 0xaa 0x0a。 The purpose is to avoid the same with the escape character 0xaa。

Receive :

Received symbol 0xa5, said that the beginning of a packet

Received symbol 0xae, said that the end of a packet

When PC receive data from controller, if there is 0xA5, 0xAA or 0xAE, it should convert two code to one code, specifically for

0xaa 0x05 ➔ 0xa5

0xaa 0x0e ➔ 0xae

0xaa 0x0a ➔ 0xaa

2.2 Network data packet format

2.2.1 The data packet format of network sending

Data	Value	Length(Byte)	Description
ID Code	0x00000000 ~ 0xffffffff	4	Control network ID code, high byte in the former。

			Need to set to the same value on the card.
Network data length	0x0000 ~ 0xffff	2	The byte length that from "Packet type" to "Packet data checksum".
Reservation	0x0000	2	Reservations. Fill in 0
Packet type	0x68	1	Recognition of this type of packet
Card type	0x32	1	Fixed Type Code
Card ID	0x01~0xFE 0XFF	1	Control card ID, the screen No, valid values are as follows: 1 ~ 254: the specified card ID 0XFF: that group address, unconditionally receiving data
Command code (CMD)		1	To perform the specified command
Additional information/ confirmation mark	FF	1	The meaning of bytes in the packet is sent, "Additional Information", is a packet plus instructions, and now only use the lowest: bit 0: whether to return a confirmation, 1 to return and 0 not to return bit1 ~ bi7: reserved, set to 0
Packet data	CC。。。。。。	Variable-length	Data
Packet data checksum	0x0000~0xffff	2	Two bytes, checksum。 Lower byte in the former。The sum of each byte from " Packet type " to " Packet data" content。

Data within the network packet does not need to do the transcoding process。

2.2.2 The data packet format of the control card returned to network sender

Data	Value	Length(Byte)	Description
ID Code	0x00000000 ~ 0xffffffff	4	Control network ID code, high byte in the former。 Need to set to the same value on the

			card.
Network data length	0x0000 ~ 0xffff	2	The byte length that from "Packet type" to "Packet data checksum".
Reservation	0x0000	2	Reservations. Fill in 0
Packet type	0xE8	1	Recognition of this type of packet 0xE8 = (0x68 | 0x80), for the app 3.2 or below return 0x68, app 3.3 or above return 0xe8 to same as other protocol(such as "set time" protocol), so you can ignore the highest bit (0x80), then it works for all app version
Card type	0x32	1	Fixed Type Code
Card ID	0x01~0xFE 0XFF	1	Control card ID, the screen No, valid values are as follows: 1 ~ 254: the specified card ID 0XFF: that group address, unconditionally
Command code (CMD)		1	With the packets sent
Return value	RR	1	RR = 0x00: that successful; RR = 0x01 ~ 0xFF: that the failure error code. In addition, a certain period of time does not receive the returned data packet, said communication failures.
Packet data	CC 。。。。。。	Variable-length	Data
Packet data checksum	0x0000~0xffff	2	Two bytes, checksum。 Lower byte in the former。The sum of each byte from " Packet type " to " Packet data" content 。

The network packet data does not need to do transcoding processing.

2.3 Command list

Command	Command sub-code	Description
Query and set network parameter	0x3c	
Query and set ID、baud rate	0x3e	

2.4 Data format

2.4.1 Query and set network parameter (CMD = 0x3c)

2.4.1.1 Query network parameter

Send packet：

Data Position	Data Items	Length(Byte)	Description
0x0000	1	1	1: Query network parameter

Return packet：

Data Position	Data Items	Length(Byte)	Description
0x0000	Confirmation message	1	0 Failed；1 Successful.
0x0001	IP Address	4	IP Address
0x0005	Gateway	4	Gateway
0x0009	Subnet mask	4	Subnet mask
0x000d	IP port number	2	IP port number
0x000f	Network ID code	4	Network ID code

2.4.1.2 Set network parameter

Send packet：

Data Position	Data Items	Length(Byte)	Description
0x0000	0	1	0: Set network parameter
0x0001	IP Address	4	IP Address
0x0005	Gateway	4	Gateway
0x0009	Subnet mask	4	Subnet mask
0x000d	IP port number	2	IP port number
0x000f	Network ID code	4	Network ID code

Return packet：

Data Position	Data Items	Length(Byte)	Description
0x0000	Confirmation message	1	0 Failed； 1 Successful.

Serial data example：(IP=192.168.1.222)
A5 68 32 01 3C 01 00 C0 A8 01 DE C0 A8 01 01 FF FF FF 00 14 50 FF FF FF FF E6 0B AE
A5 E8 32 01 3C 01 01 59 01 AE

2.4.2 Query and set ID and baud rate (CMD = 0x3e)

2.4.2.1 Query controller ID and baud rate

Send packet：

Data Position	Data Items	Length(Byte)	Description
0x0000	1	1	1: Query controller ID and baud rate
0x0001	0	1	Reservation
0x0002	0	1	Reservation

Return packet：

Data Position	Data Items	Length(Byte)	Description
0x0000	Confirmation message	1	0 Failed ; 1 Successful.
0x0001	ID number	1	ID number
0x0002	Baud rate number	1	0: 115200 1: 57600 2: 38400 3: 19200 4: 9600 5: 4800 6: 2400

2.4.2.2 Set controller ID and baud rate

Send packet：

Data Position	Data Items	Length(Byte)	Description
0x0000	0	1	0: Set controller ID and baud rate
0x0001	ID number	1	1~254
0x0002	Baud rate number	1	0: 115200 1: 57600 2: 38400 3: 19200 4: 9600 5: 4800 6: 2400

Return packet：

Data Position	Data Items	Length(Byte)	Description
0x0000	Confirmation message	1	0 Failed ; 1 Successful.